Lab Exercise 15- Creating ConfigMaps in Kubernetes

In this lab exercise, you will learn how to create ConfigMaps in Kubernetes using different methods: from literal values, from a file, from an environment file, and from multiple files. ConfigMaps allow you to decouple configuration artifacts from image content to keep containerized applications portable.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

Step 2: Create a ConfigMap from Literal Values

Create a ConfigMap named example-configmap with literal values:

```
kubectl create configmap example-configmap --from-literal=key1=value1 --from-literal=key2=value2
```

Verify the ConfigMap:

```
kubectl get configmap example-configmap -o yaml
```

You should see the example-configmap with the keys and values specified.

Step 3: Create a ConfigMap from a File

Create a file named **config.txt** with the following content:

```
key1=value1
key2=value2
```

Create a ConfigMap named file-configmap from the file:

```
kubectl create configmap file-configmap --from-file=config.txt
```

Verify the ConfigMap:

```
kubectl get configmap file-configmap -o yaml
```

You should see the file-configmap with the content of config.txt.

Step 4: Create a ConfigMap from an Environment File

Create an environment file named **env-config.env** with the following content:

```
ENV_VAR1=value1
ENV_VAR2=value2
```

Create a ConfigMap named env-configmap from the environment file:

```
kubectl create configmap env-configmap --from-env-file=env-config.env
```

Verify the ConfigMap:

```
kubectl get configmap env-configmap -o yaml
```

You should see the env-configmap with the environment variables specified in the file.

Step 5: Create a ConfigMap from Multiple Files

Create multiple files:

config1.txt with content:

```
key1=value1
```

config2.txt with content:

```
key2=value2
```

Create a ConfigMap named multi-file-configmap from multiple files:

```
kubectl create configmap multi-file-configmap --from-file=config1.txt --from-file=config2.txt
```

Verify the ConfigMap:

```
kubectl get configmap multi-file-configmap -o yaml
```

You should see the multi-file-configmap with the content of both config1.txt and config2.txt.

Step 6: Use ConfigMaps in a Pod

Create a file named **pod-configmap.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
        name: env-configmap
```

In this manifest:

- The Pod reads environment variables from the env-configmap.
- The Pod mounts the file-configmap at /etc/config.

Apply the manifest to create the Pod:

```
kubectl apply -f pod-configmap.yaml
```

Check the status of the Pod:

```
kubectl get pods
```

View the Pod logs to see the environment variables and file content:

```
kubectl exec -it configmap-pod -- printenv
```

You should see the environment variables and the content of the file-configmap.

Step 7: Clean Up

After completing the exercise, clean up the resources created:

```
kubectl delete pod configmap-pod
kubectl delete configmap example-configmap
kubectl delete configmap file-configmap
kubectl delete configmap env-configmap
kubectl delete configmap multi-file-configmap
rm config.txt env-config.env config1.txt config2.txt
```

```
C:\Users\NK>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


C:\Users\NK>minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home 25H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Updating the running docker "minikube" container ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

C:\Users\NK>kubectl create configmap example-configmap --from-literal=key1=value1 --from-literal=key2=value2
configmap/example-configmap created

C:\Users\NK>kubectl get configmap example-configmap -o yaml
apiVersion: v1
data:
  key1: value1
  key2: value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-04-28T04:10:42Z"
  name: example-configmap
  namespace: default
  resourceVersion: "3561"
  uid: bf938bc6-db3c-4cff-9ad3-aa13af039c30

C:\Users\NK>(echo key1=value1 & echo key2=value2) > config.txt
```

```
configmap/multi-file-configmap created

C:\Users\NK>kubectl get configmap multi-file-configmap -o yaml
apiVersion: v1
data:
  config1.txt: "key1=value1 \r\n"
  config2.txt: "key2=value2 \r\n"
kind: ConfigMap
metadata:
  creationTimestamp: "2026-04-28T04:11:24Z"
  name: multi-file-configmap
  namespace: default
  resourceVersion: "3603"
  uid: bf0c7b64-6602-4607-8ba3-900caafce88a

C:\Users\NK>(echo apiVersion: v1 & echo kind: Pod & echo metadata: & echo   name: configmap-pod & echo spec: & echo   containers: & echo   - name: nginx & e
cho     image: nginx & echo     envFrom: & echo     - configMapRef: & echo         name: env-configmap) > pod-configmap.yaml

C:\Users\NK>kubectl apply -f pod-configmap.yaml
pod/configmap-pod created

C:\Users\NK>kubectl get pods
NAME           READY   STATUS              RESTARTS      AGE
configmap-pod   0/1     ContainerCreating   0             2s
my-pod          1/1     Running             1 (78s ago)   55m

C:\Users\NK>kubectl exec -it configmap-pod -- printenv
error: Internal error occurred: unable to upgrade connection: container not found ("nginx")

C:\Users\NK>kubectl delete pod configmap-pod
pod "configmap-pod" deleted from default namespace

C:\Users\NK>kubectl delete configmap example-configmap
configmap "example-configmap" deleted from default namespace

C:\Users\NK>kubectl delete configmap file-configmap
configmap "file-configmap" deleted from default namespace

C:\Users\NK>kubectl delete configmap env-configmap
Error from server (NotFound): configmaps "env-configmap" not found
```

```
C:\Users\NK>kubectl delete configmap file-configmap
configmap "file-configmap" deleted from default namespace

C:\Users\NK>kubectl delete configmap env-configmap
Error from server (NotFound): configmaps "env-configmap" not found

C:\Users\NK>kubectl delete configmap multi-file-configmap
configmap "multi-file-configmap" deleted from default namespace

C:\Users\NK>del config.txt env-config.env config1.txt config2.txt

C:\Users\NK>kubectl get configmap
NAME                DATA   AGE
kube-root-ca.crt    1      55m

C:\Users\NK>
```